

# CS383 - Software Engineering

## Introduction (I)

## Topics to be covered in this lecture

- Provide the course profile and grading details
- Group project and assessment
- Software disasters
- Software FAQs
- Motivation of SE
- Important of SE

# Course profile

- Overview

a course to teach you how software is developed systematically using best development and engineering practices

- Module syllabus:

- Course syllabus is available on the blackboard (please check it).

- Text Books:

- I. Sommerville, "Software Engineering", tenth edition.
- Martin Fowler, "UML Distilled", third edition
- Gang of Four (E. Gamma, R. Helm, R. Johnson & J. Vlissides) "Design Patterns", 1994

- Assessment and grading:

- 1 midterm exam = %20 (Week 7 second session for all sections)
- Group project = 25%+5% (assessment is undertaking throughout the course and the project presentation take place in week 14)
- Final exam = %50

# Group Project Assignment

- You'll all be allocated into groups
- Once allocated, you need to come up with an idea for system/software
  - There is no implementation/coding required
  - Idea must be well-motivated, innovative and well-inspired
- You need to start developing and apply what you are learning during lectures into your development process
- You need regular meetings for briefing of the development process. You are required to organize that (8 meetings at least)
- Individual assessment is applied and your contribution is considered to determine your final grade
- See for full description about the project and points/criteria of assessment.

## Group Project Assignment (cont.)

- You must please have a leader: vote wisely for ONE
- You need to demonstrate aspects of SWE into your project:
  - Project management, tasks allocation, and teamwork
  - Group meeting documentations
    - separate docs for agenda and minutes
  - Software documentation (SRS, SDD, SE ethics)
  - What development process you have chosen and why
  - Address functional and non-functional requirements
  - System/software design
  - Use UML diagrams and modeling tools
  - Presentation
- You should participate in each task of the above

# Software Costs

- Knight's \$440 Million Error
  - Stocks struggled to stay afloat after a software bug triggered a \$440 million loss in just 30 minutes. [1]
  
- NASA's Mars Climate Orbiter (1998)
  - The spacecraft (\$125 million) was ultimately lost in space.
  - A sub contractor on the engineering team failed to make a simple conversion from imperial units to metric systems. [2]



[1] <https://archive.nytimes.com/dealbook.nytimes.com/2012/08/02/knight-capital-says-trading-mishap-cost-it-440-million>

[2] <https://www.simscale.com/blog/nasa-mars-climate-orbiter-metric/>

## Software Costs (cont.d)

- Colonial pipeline's attack (2021) [3]
  - A stolen access credentials used to access a VPN of the company
  - No multifactor authentication is applied
  - Damage:
    - Pipeline shutdown: halt all fuel delivery
    - 100+ gigabytes of data were stolen
    - Ransom: \$5 million



## Software Costs (cont.)

- Software costs often dominate computer system costs
  - The costs of software on a PC are often greater than the hardware cost
  
- Software costs (time-wise) to be maintained more than to be developed
  - Development costs 60%
  - Testing costs 40%
  - Maintenance and software evolution costs more than development and testing





## Software Costs (cont.)



So, what is software engineering?

SE is a discipline that is concerned with all aspects of producing cost-effective and trustworthy software

## FAQ about SE

### **Q. What is software?**

A computer program and associated documentation.

### **Q. What are the attributes of good software?**

Good software should deliver the demanded functional and nonfunctional requirements of users/clients/customers in most efficient and effective manner. Also, software should be maintainable, dependable and usable.

## FAQ about SE

### **Q. What are the fundamental software engineering activities?**

Software specification, software development, software verification & validation, software evolution. Software development is two-fold parts: software design and the implementation of the design

### **Q. What is the difference between Software Engineering and system engineering?**

System Engineering is more general: concerned with system development including hardware, software and engineering process.

Software engineering is more specific about software development

## FAQ about SE

### **Q. What are the best SE techniques and methods?**

It depends. E.g. games development goes under series of prototypes to receive early feedback

Whereas safety critical control systems require analyzable specification and documentation and design models to be developed

# Software Product

## 1. Generic products

Stand-alone systems that are marketed and sold to any customer



Examples – PC software like Microsoft Office, CAD software; appointment system for dentists, etc.



**Final Cut Pro X**

# Software Product (cont.)

## 2. Customized Product

Software that is commissioned (tailored) by specific clients to meet their own customer needs

Examples – embedded control systems,  
air traffic control software, traffic monitoring systems, etc.



# Motivation behind SE



Software development comes with great challenges. Namely, how to:

- Manage Large software.
- Produce scalable software.
- Handle high cost of development and maintenance
- Manage the dynamic nature of software.
- Produce better quality software

# Why SE is important?



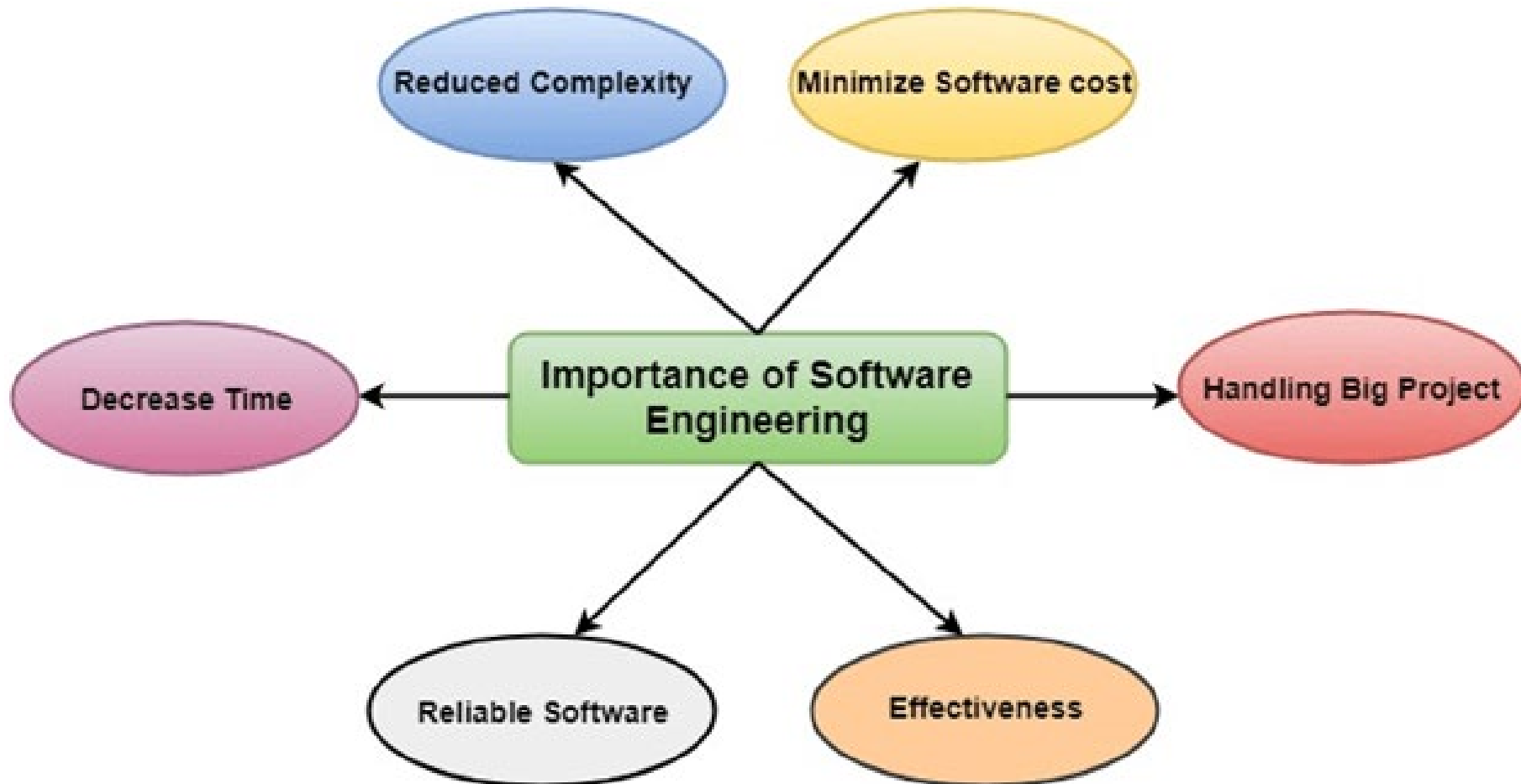
- **Reduces complexity:** software engineering divides big problems into small sub-problems. All these small problems are solved, programmed and tested independently.
- **To minimize software cost:** the cost for software productions becomes less as compared to any software that does not use software engineering method.
- **To decrease time of development:** if you are making your software according to the software engineering method and plan, then this will decrease huge time of development.



## Why SE is important? (cont.)

- **Handling big projects:** big projects require heap of planning, direction, testing, and maintenance. So to handle a big project without any problem, project should be development according to a plan.
- **Reliable software:** in software engineering, testing and maintenance are given, so the confidence about the software reliability is achieved.
- **Effectiveness:** effectiveness comes if anything has made according to the standards. Software standards are the big target of companies to make it more effective. So Software becomes more effective in the act with the help of software engineering.

# SE importance summary



# Issues related to SE



## Heterogeneity

Systems are increasingly essential to operate as distributed systems across networks, which include various types of computers and mobile devices.



## Business and social change

Business and society are evolving rapidly. They need to be in a position to update their current software and create new software quickly.

# Issues related to SE



## Security and trust

We must ensure that unauthorized users are unable to target our applications, and that information is protected and secured.



## Scales

The software must be built over a broad range of scales, from very tiny embedded systems in portable or wearable devices through the Internet sizes, a cloud-based framework that serves global population.

# Application types



Software engineering diversity:

- There are many different types of software system and there is no universal set of software techniques that is applicable to all of these.
- The software engineering methods and tools used for development depend on:
  - The type of application being developed,
  - The requirements of the customer,
  - The background of the development team (people involved),
  - Tools used for development and software engineering